

1 Gradient Descent

$$C'(w) = \lim_{\epsilon \rightarrow 0} \frac{C(w + \epsilon) - C(w)}{\epsilon} \quad (1)$$

(2)

1.1 “Twice”

$$C(w) = \frac{1}{n} \sum_{i=1}^n (x_i w - y_i)^2 \quad (3)$$

$$C'(w) = \left(\frac{1}{n} \sum_{i=1}^n (x_i w - y_i)^2 \right)' \quad (4)$$

$$= \frac{1}{n} \left(\sum_{i=1}^n (x_i w - y_i)^2 \right)' \quad (5)$$

$$= \frac{1}{n} \sum_{i=1}^n ((x_i w - y_i)^2)' \quad (6)$$

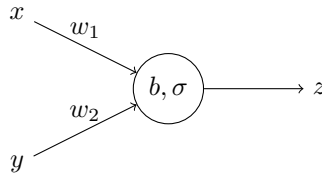
$$= \frac{1}{n} \sum_{i=1}^n 2(x_i w - y_i) (x_i w - y_i)' \quad (7)$$

$$= \frac{1}{n} \sum_{i=1}^n 2(x_i w - y_i) x_i \quad (8)$$

$$C(w) = \frac{1}{n} \sum_{i=1}^n (x_i w - y_i)^2 \quad (9)$$

$$C'(w) = \frac{1}{n} \sum_{i=1}^n 2(x_i w - y_i) x_i \quad (10)$$

1.2 One Neuron Model with 2 inputs



$$y = \sigma(xw_1 + yw_2 + b) \quad (11)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (12)$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)) \quad (13)$$

1.2.1 Cost

$$a_i = \sigma(x_iw_1 + y_iw_2 + b) \quad (14)$$

$$\partial_{w_1} a_i = a_i(1 - a_i)\partial_{w_1}(x_iw_1 + y_iw_2 + b) \quad (15)$$

$$= a_i(1 - a_i)x_i \quad (16)$$

$$\partial_{w_2} a_i = a_i(1 - a_i)y_i \quad (17)$$

$$\partial_b a_i = a_i(1 - a_i) \quad (18)$$

$$C = \frac{1}{n} \sum_{i=1}^n (a_i - z_i)^2 \quad (19)$$

$$\partial_{w_1} C = \frac{1}{n} \sum_{i=1}^n \partial_{w_1} ((a_i - z_i)^2) \quad (20)$$

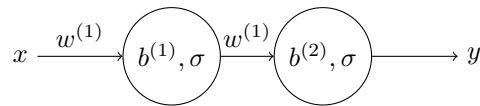
$$= \frac{1}{n} \sum_{i=1}^n 2(a_i - z_i)\partial_{w_1}(a_i) \quad (21)$$

$$\partial_{w_1} C = \frac{1}{n} \sum_{i=1}^n 2(a_i - z_i)a_i(1 - a_i)x_i \quad (22)$$

$$\partial_{w_2} C = \frac{1}{n} \sum_{i=1}^n 2(a_i - z_i)a_i(1 - a_i)y_i \quad (23)$$

$$\partial_b C = \frac{1}{n} \sum_{i=1}^n 2(a_i - z_i)a_i(1 - a_i) \quad (24)$$

1.3 Two Neurons Model with 1 input



$$a^{(1)} = \sigma(xw^{(1)} + b^{(1)}) \quad (25)$$

$$y = \sigma(a^{(1)}w^{(2)} + b^{(2)}) \quad (26)$$

$$(27)$$

1.3.1 Cost

$$a_i^{(1)} = \sigma(x_i w^{(1)} + b^{(1)}) \quad (28)$$

$$a_i^{(2)} = \sigma(a_i^{(1)} w^{(2)} + b^{(2)}) \quad (29)$$

$$\partial_{w^{(2)}} a_i^{(2)} = a_i^{(2)} (1 - a_i^{(2)}) a_i^{(1)} \quad (30)$$

$$\partial_{b^{(2)}} a_i^{(2)} = a_i^{(2)} (1 - a_i^{(2)}) \quad (31)$$

$$\partial_{a_i^{(1)}} a_i^{(2)} = a_i^{(2)} (1 - a_i^{(2)}) w^{(2)} \quad (32)$$

$$C^{(2)} = \frac{1}{n} \sum_{i=1}^n (a_i^{(2)} - y_i)^2 \quad (33)$$

$$\partial_{w^{(2)}} C^{(2)} = \frac{1}{n} \sum_{i=1}^n \partial_{w^{(2)}} \left((a_i^{(2)} - y_i)^2 \right) \quad (34)$$

$$= \frac{1}{n} \sum_{i=1}^n 2(a_i^{(2)} - y_i) \partial_{w^{(2)}} \left(a_i^{(2)} \right) \quad (35)$$

$$= \frac{1}{n} \sum_{i=1}^n 2(a_i^{(2)} - y_i) a_i^{(2)} (1 - a_i^{(2)}) a_i^{(1)} \quad (36)$$

$$\partial_{b^{(2)}} C^{(2)} = \frac{1}{n} \sum_{i=1}^n 2(a_i^{(2)} - y_i) a_i^{(2)} (1 - a_i^{(2)}) \quad (37)$$

$$\partial_{a_i^{(1)}} C^{(2)} = \frac{1}{n} \sum_{i=1}^n 2(a_i^{(2)} - y_i) a_i^{(2)} (1 - a_i^{(2)}) w^{(2)} \quad (38)$$

$$(39)$$

using the chain rule that:

$$\partial_{w^{(1)}} C^{(2)} = \frac{1}{n} \sum_{i=1}^n \partial_{a_i^{(1)}} C^{(2)} \cdot \partial_{w^{(1)}} a_i^{(1)} \quad (40)$$

$$= \frac{1}{n} \sum_{i=1}^n \partial_{a_i^{(2)}} C^{(2)} \cdot \partial_{a_i^{(1)}} a_i^{(2)} \cdot \partial_{w^{(1)}} a_i^{(1)} \quad (41)$$

so we can use the $\partial_{a_i^{(1)}} C^{(2)}$ to move backwards.

but before that, we need to define the “Cost” of the first layer

$$a_i^{(1)} = \sigma(x_i w^{(1)} + b^{(1)}) \quad (42)$$

$$\partial_{w^{(1)}} a_i^{(1)} = a_i^{(1)}(1 - a_i^{(1)})w^{(1)} \quad (43)$$

$$e_i = a_i^{(1)} - \partial_{a_i^{(1)}} C^{(2)} \quad (44)$$

$$C^{(1)} = \frac{1}{n} \sum_{i=1}^n (a_i^{(1)} - e_i)^2 \quad (45)$$

$$\partial_{w^{(1)}} C^{(1)} = \partial_{w^{(1)}} \left(\frac{1}{n} \sum_{i=1}^n (a_i^{(1)} - e_i)^2 \right) \quad (46)$$

$$= \frac{1}{n} \sum_{i=1}^n \partial_{w^{(1)}} \left((a_i^{(1)} - e_i)^2 \right) \quad (47)$$

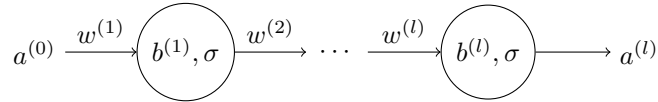
$$= \frac{1}{n} \sum_{i=1}^n 2(a_i^{(1)} - e_i) \partial_{w^{(1)}} (a_i^{(1)}) \quad (48)$$

$$= \frac{1}{n} \sum_{i=1}^n 2(\partial_{a_i^{(1)}} C^{(2)}) x_i \quad (49)$$

$$\partial_{b^{(1)}} C^{(1)} = \frac{1}{n} \sum_{i=1}^n 2(\partial_{a_i^{(1)}} C^{(2)}) \quad (50)$$

note that this is literally the same as to make it out with chain rule, but it's easier to code in this way.

1.4 Arbitrary Neurons Model with 1 input



1.4.1 Feed-Forward

$$a_i^{(j)} = \sigma(a_i^{(j-1)} w^{(j)} + b^{(j)}) \quad (51)$$

$$\partial_{w^{(j)}} a_i^{(j)} = a_i^{(j)}(1 - a_i^{(j)})a_i^{(j-1)} \quad (52)$$

$$\partial_{b^{(j)}} a_i^{(j)} = a_i^{(j)}(1 - a_i^{(j)}) \quad (53)$$

$$\partial_{a_i^{(j-1)}} a_i^{(j)} = a_i^{(j)}(1 - a_i^{(j)})w^{(j)} \quad (54)$$

1.4.2 Back-Propagation

$$e_i^{(j)} = a_i^{(j)} - \partial_{a_i^{(j)}} C^{(j+1)} \quad (55)$$

$$C^{(j)} = \frac{1}{n} \sum_{i=1}^n (a_i^{(j)} - e_i^{(j)})^2 \quad (56)$$

$$\partial_{w^{(j)}} C^{(j)} = \partial_{w^{(j)}} \left(\frac{1}{n} \sum_{i=1}^n (a_i^{(j)} - e_i^{(j)})^2 \right) \quad (57)$$

$$= \frac{1}{n} \sum_{i=1}^n \partial_{w^{(j)}} \left((a_i^{(j)} - e_i^{(j)})^2 \right) \quad (58)$$

$$= \frac{1}{n} \sum_{i=1}^n 2(a_i^{(j)} - e_i^{(j)}) \partial_{w^{(j)}} (a_i^{(j)}) \quad (59)$$

$$= \frac{1}{n} \sum_{i=1}^n 2(\partial_{a_i^{(j)}} C^{(j+1)}) a_i^{(j)} (1 - a_i^{(j)}) a_i^{(j-1)} \quad (60)$$

$$\partial_{w^{(j)}} C^{(j)} = \frac{1}{n} \sum_{i=1}^n 2(\partial_{a_i^{(j)}} C^{(j+1)}) a_i^{(j)} (1 - a_i^{(j)}) \quad (61)$$

$$\partial_{a_i^{(j-1)}} C^{(j)} = \frac{1}{n} \sum_{i=1}^n 2(\partial_{a_i^{(j)}} C^{(j+1)}) a_i^{(j)} (1 - a_i^{(j)}) w^{(j)} \quad (62)$$